Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

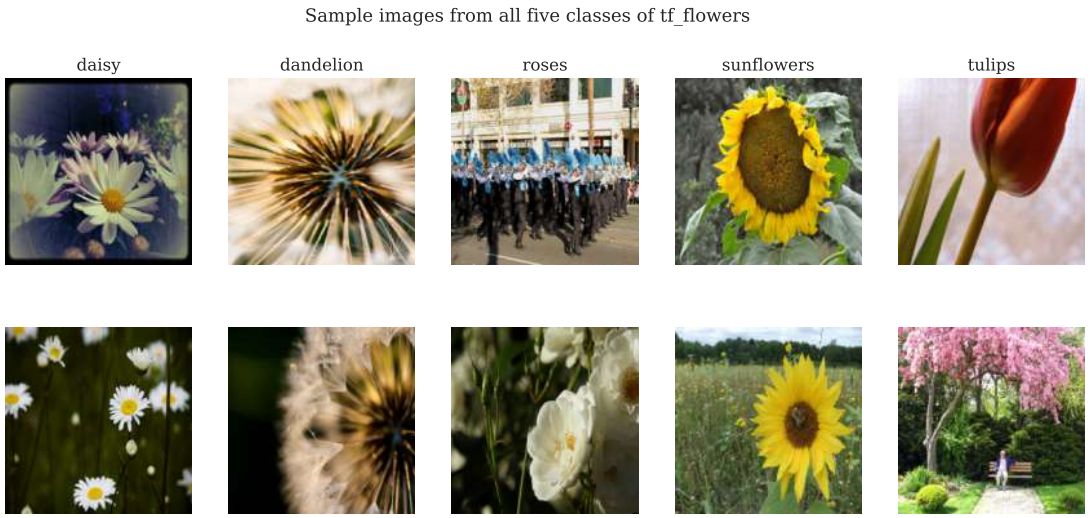
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

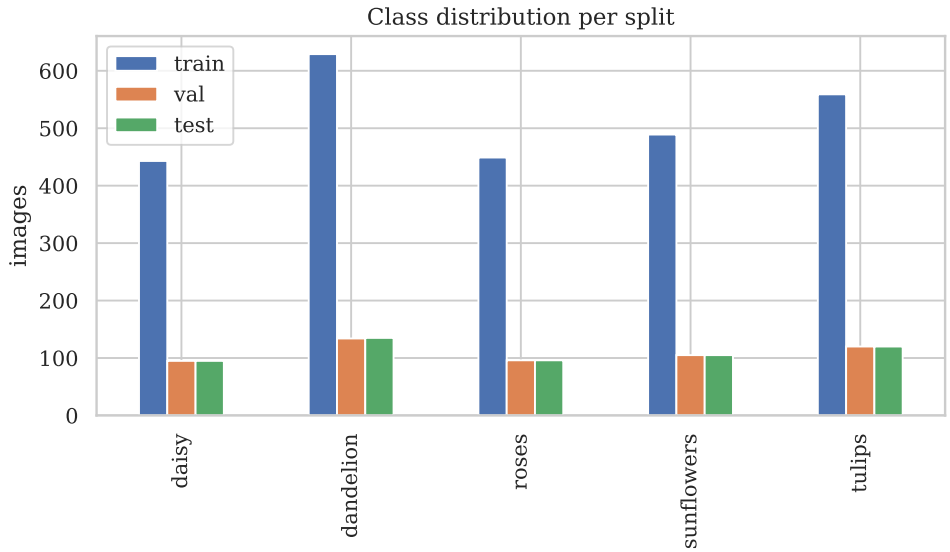
`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.





3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50. Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and

- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.

This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.

For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.

Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.

Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	**Location**	**Resolution**	**Channels**
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^{C}.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\mathrm{Conv}_{7\times7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

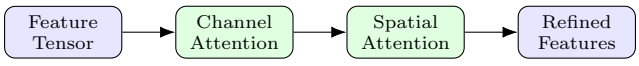
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H\times W\times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N\times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

Model	Accuracy	Precision	Recall	Macro F1	Parameters	Inference (ms)
Baseline ResNet50	94.92%	95.05%	94.81%	0.9489	23,518,277	3.0702
Channel Attention (SE)	94.74%	94.88%	94.60%	0.9471	23,650,437	3.2828
Spatial Attention ($k = 7$)	95.10%	95.14%	95.11%	0.9512	23,518,375	3.3299
Channel+Spatial (P_3)	94.19%	94.45%	94.04%	0.9414	23,650,535	3.3278
Self-Attention (P_3)	93.47%	93.45%	93.44%	0.9342	24,830,790	3.5036
Combined Attention (P_3)	94.92%	95.39%	94.67%	0.9493	24,963,048	3.5254

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

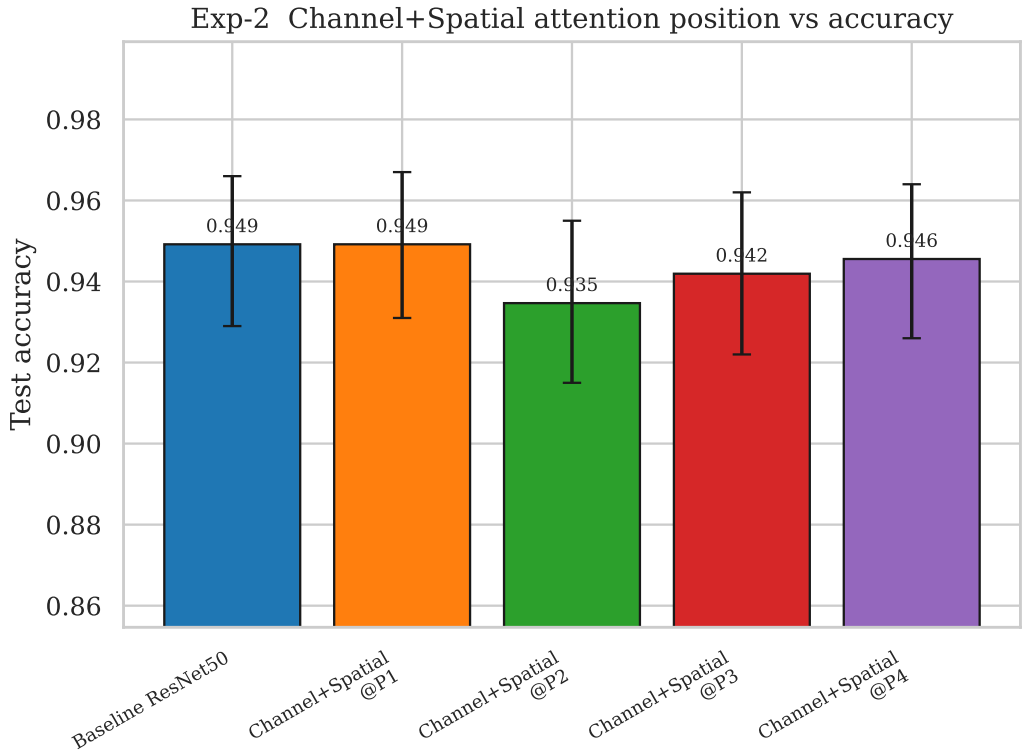
Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

How does the location of attention influence performance?

Position	Accuracy	Precision	Recall	Macro F1	Attention Params
P_1	94.92%	94.80%	94.94%	0.9485	8,562
P_2	93.47%	93.72%	93.24%	0.9339	33,410
P_3	94.19%	94.45%	94.04%	0.9414	132,258
P_4	94.56%	94.68%	94.46%	0.9454	526,562



16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

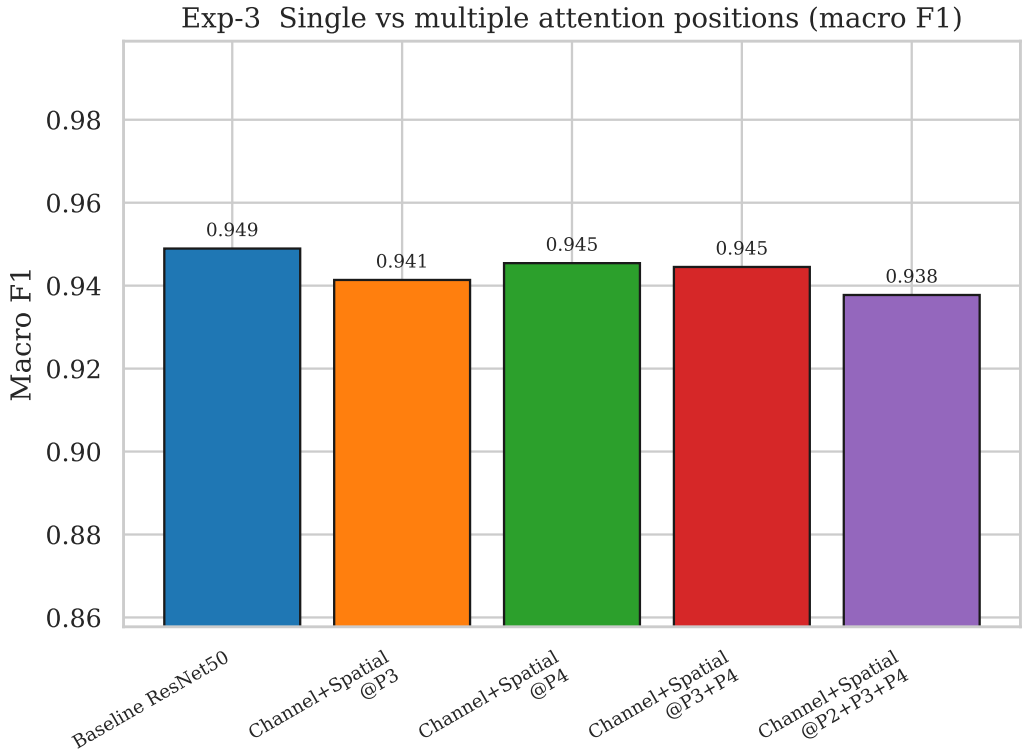
The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

Is attention at more locations necessarily more effective?

Attention Position(s)	Accuracy	Macro F1	Attention Params
P_3	94.19%	0.9414	132,258
P_4	94.56%	0.9454	526,562
$P_3 + P_4$	94.56%	0.9445	658,820
$P_2 + P_3 + P_4$	93.83%	0.9377	692,230



17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features. Compare:

Channel → Spatial

against

Spatial → Channel

while maintaining the same insertion position.
For example, at P_3:

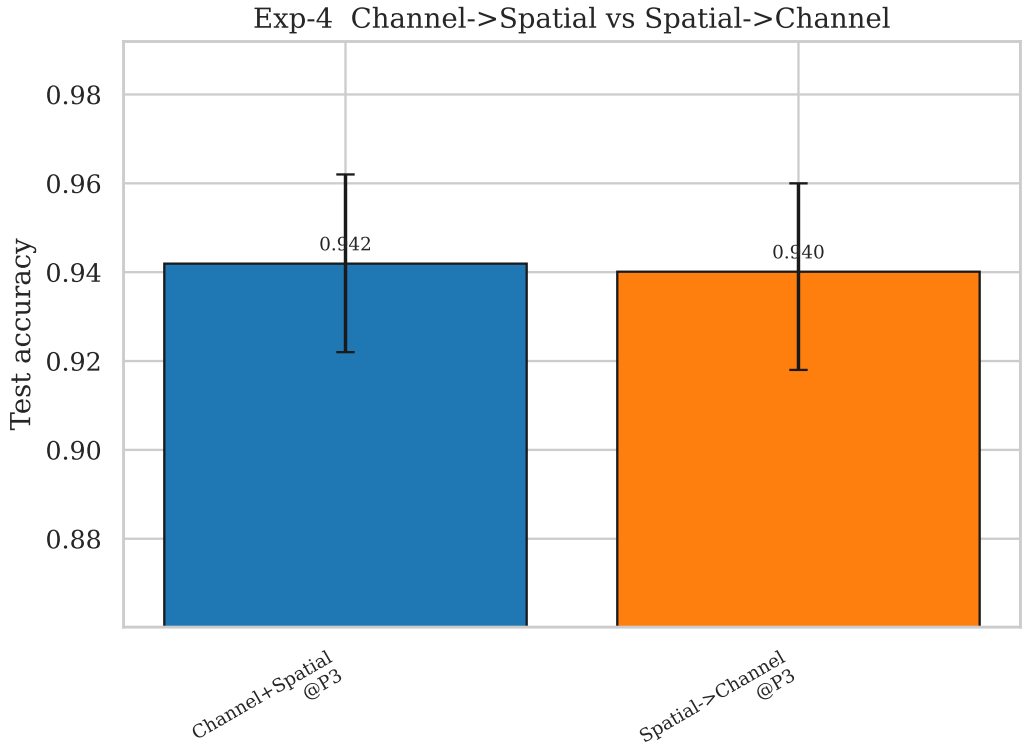
$$F \to M_c \to M_s$$

should be compared with

$$F \to M_s \to M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

Order	Accuracy	Macro F1	Inference (ms)
Channel → Spatial	94.19%	0.9414	3.3278
Spatial → Channel	94.01%	0.9394	3.3315



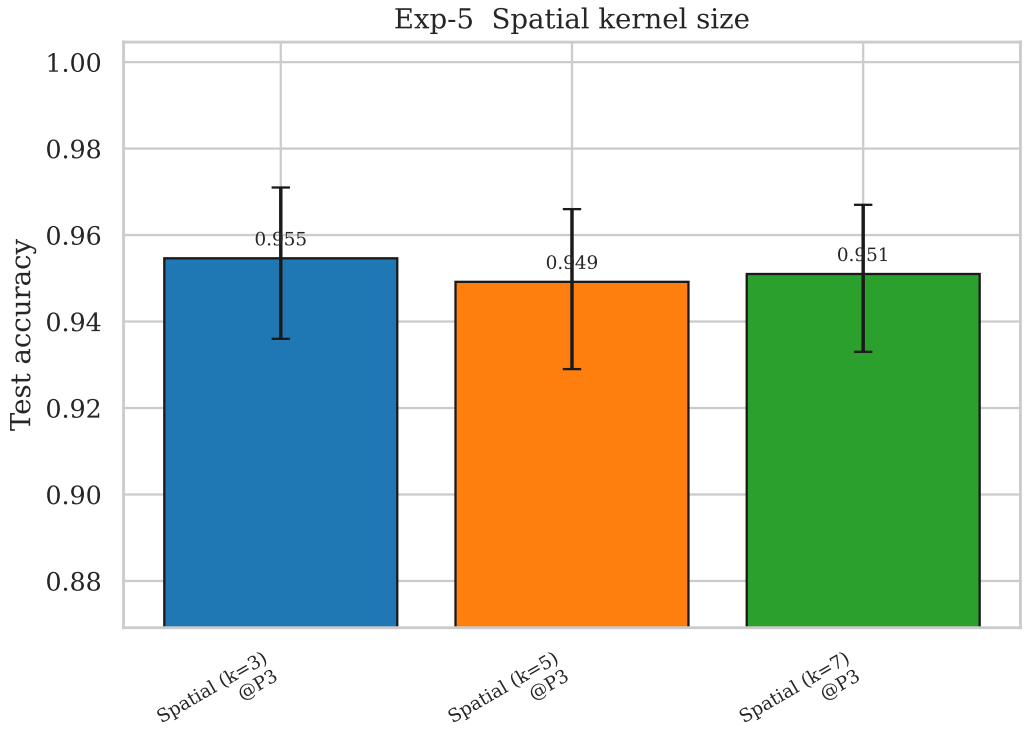
18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3	95.46%	0.9544	3.3185 ms
5×5	94.92%	0.9489	3.3297 ms
7×7	95.10%	0.9512	3.3299 ms



Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	94.92%	95.05%	94.81%	0.9489	23,518,277	3.0702 ms
Channel / SE	94.74%	94.88%	94.60%	0.9471	23,650,437	3.2828 ms
Spatial Attention	95.10%	95.14%	95.11%	0.9512	23,518,375	3.3299 ms
ECA	95.46%	95.51%	95.30%	0.9537	23,518,282	3.3348 ms
CBAM	93.28%	93.62%	93.03%	0.9320	23,650,535	3.3386 ms
Self-Attention	93.47%	93.45%	93.44%	0.9342	24,830,790	3.5036 ms
Coordinate Attention	94.56%	94.80%	94.40%	0.9452	23,618,725	3.3219 ms

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	**Position**	**Accuracy**	**Precision**	**Recall**	**Macro F1**	**Parameters**	**Inference Time**
Baseline	–	94.92%	95.05%	94.81%	0.9489	23,518,277	3.0702 ms
Channel Attention	P_3	94.74%	94.88%	94.60%	0.9471	23,650,437	3.2828 ms
Spatial Attention ($k = 7$)	P_3	95.10%	95.14%	95.11%	0.9512	23,518,375	3.3299 ms
Channel+Spatial	P_3	94.19%	94.45%	94.04%	0.9414	23,650,535	3.3278 ms
Self-Attention	P_3	93.47%	93.45%	93.44%	0.9342	24,830,790	3.5036 ms
Combined Attention	P_3	94.92%	95.39%	94.67%	0.9493	24,963,048	3.5254 ms
ECA	P_3	95.46%	95.51%	95.30%	0.9537	23,518,282	3.3348 ms
CBAM	P_3	93.28%	93.62%	93.03%	0.9320	23,650,535	3.3386 ms
Coordinate Attention	P_3	94.56%	94.80%	94.40%	0.9452	23,618,725	3.3219 ms

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

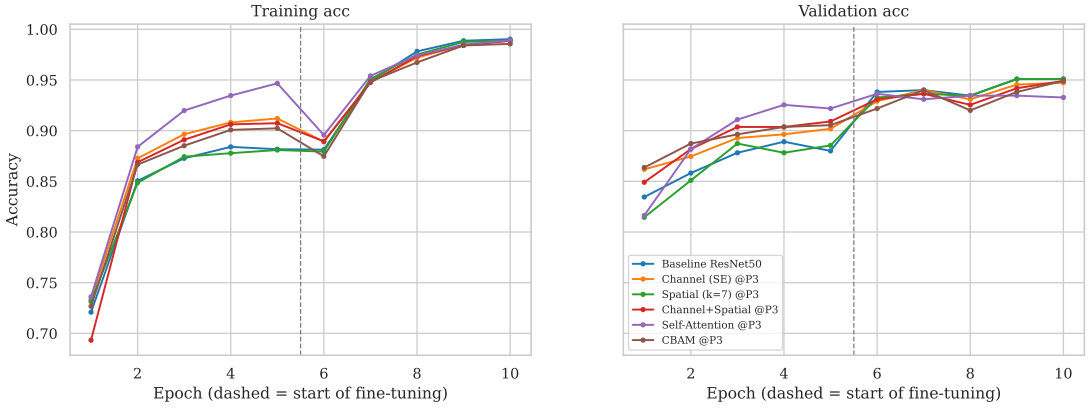
for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.1 Training / validation accuracy: baseline vs attention models



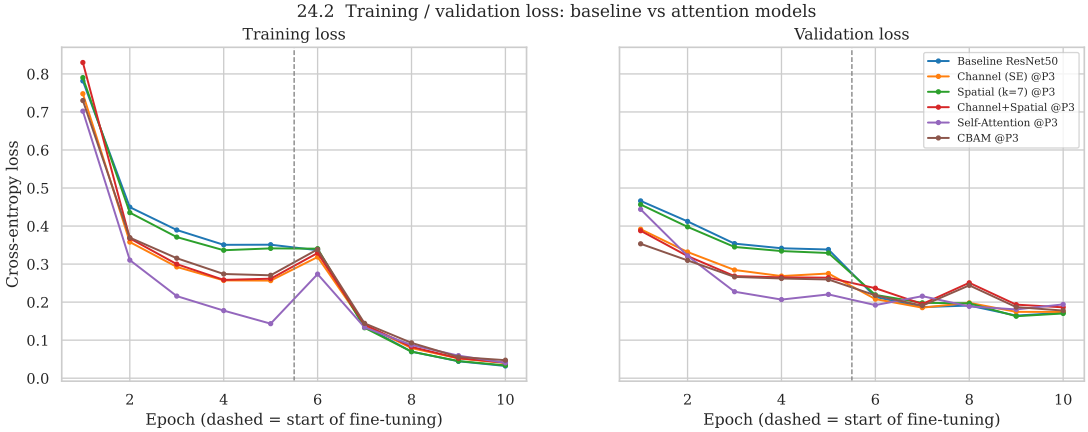
24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

24.2 Training / validation loss: baseline vs attention models



24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

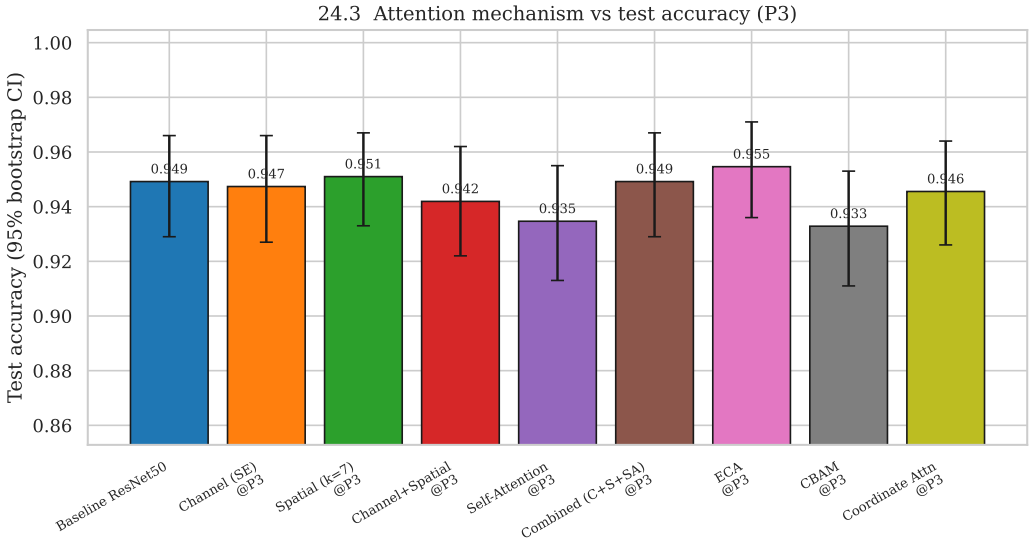
$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.



24.4 Attention Mechanism versus Macro F1

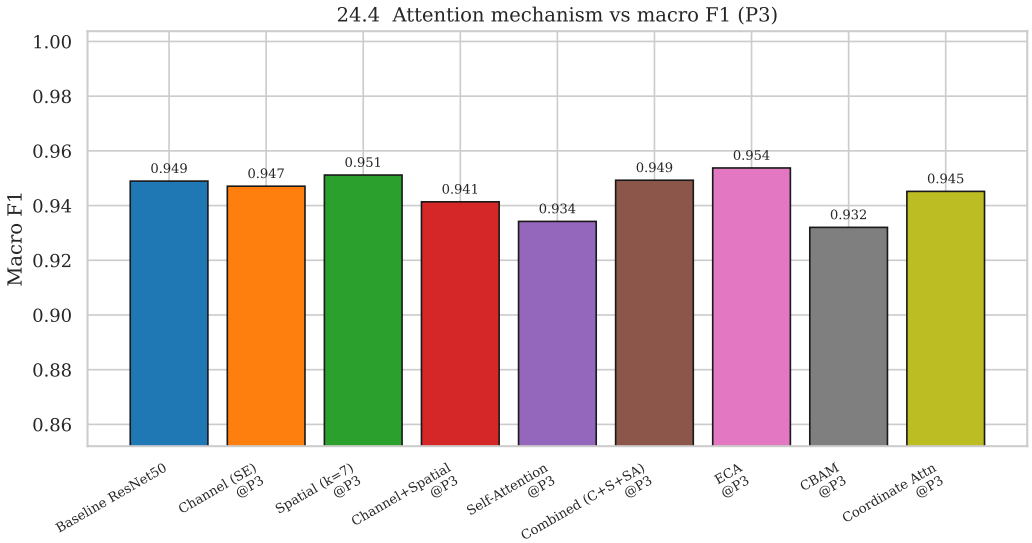
Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.



24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

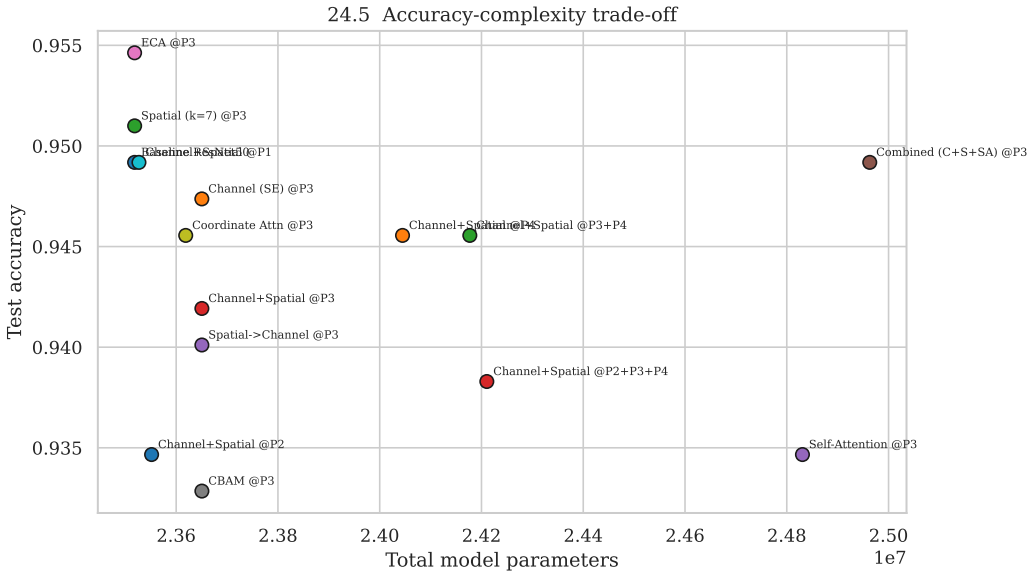
and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.



24.6 Macro F1 versus Inference Time

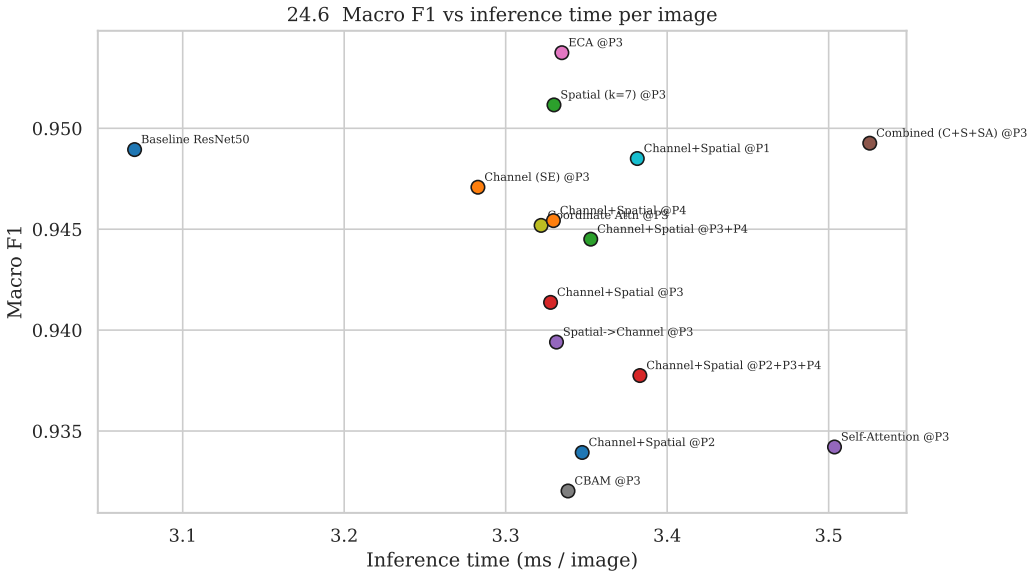
Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score}.$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.



25. Confusion Matrix Analysis

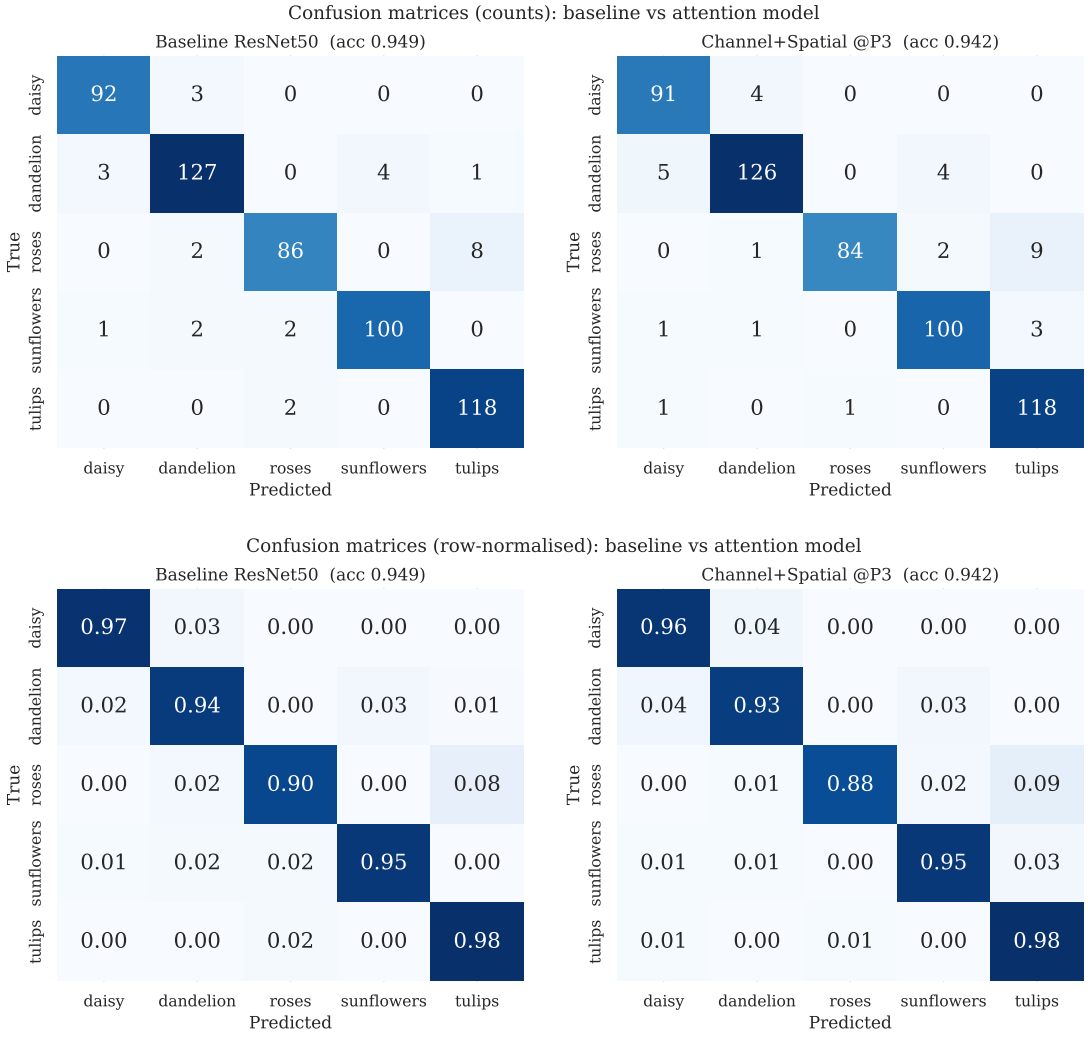
Generate a confusion matrix for:

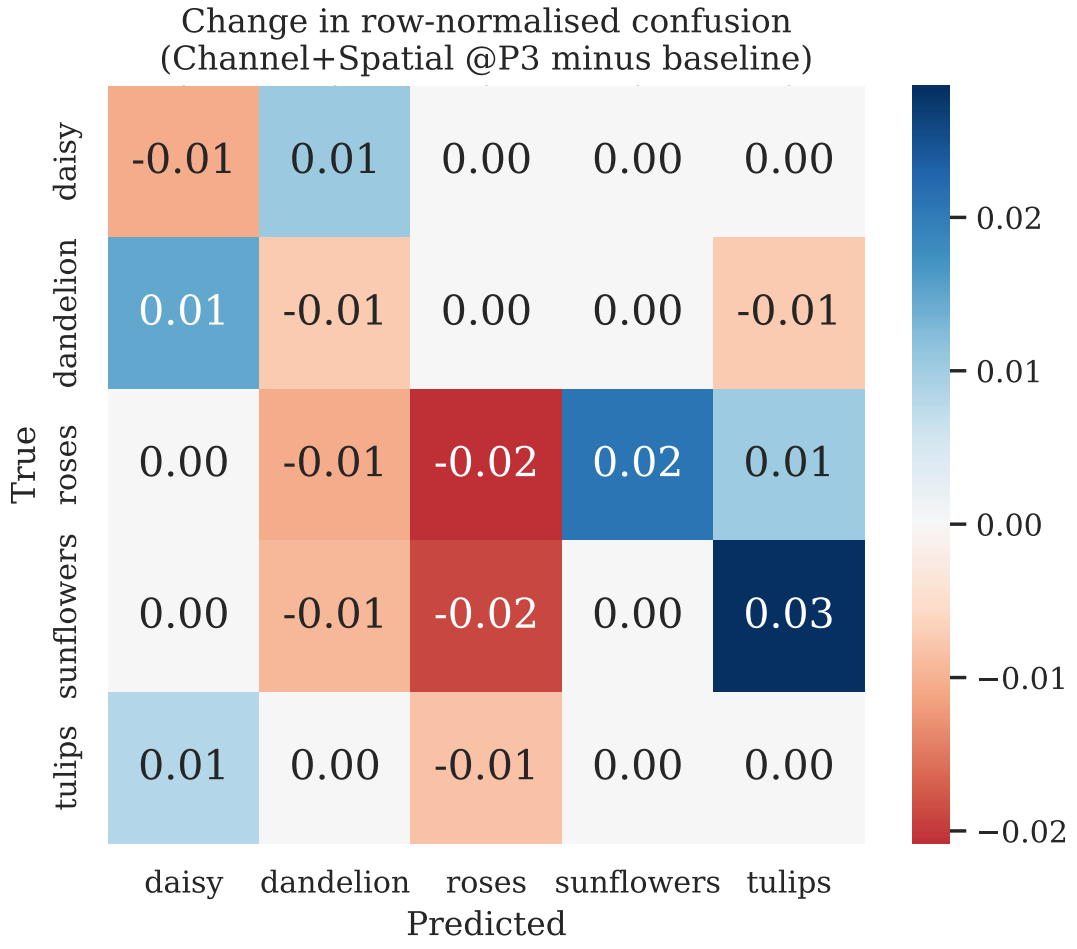
1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.


Confusion matrices (counts): baseline vs attention model
Baseline ResNet50 (acc 0.949)
Channel+Spatial @P3 (acc 0.942)
True
Predicted
daisy
dandelion
roses
sunflowers
tulips
92 3 0 0 0
3 127 0 4 1
0 2 86 0 8
1 2 2 100 0
0 0 2 0 118
91 4 0 0 0
5 126 0 4 0
0 1 84 2 9
1 1 0 100 3
1 0 1 0 118
Confusion matrices (row-normalised): baseline vs attention model
Baseline ResNet50 (acc 0.949)
Channel+Spatial @P3 (acc 0.942)
0.97 0.03 0.00 0.00 0.00
0.02 0.94 0.00 0.03 0.01
0.00 0.02 0.90 0.00 0.08
0.01 0.02 0.02 0.95 0.00
0.00 0.00 0.02 0.00 0.98
0.96 0.04 0.00 0.00 0.00
0.04 0.93 0.00 0.03 0.00
0.00 0.01 0.88 0.02 0.09
0.01 0.01 0.00 0.95 0.03
0.01 0.00 0.01 0.00 0.98


Change in row-normalised confusion
(Channel+Spatial @P3 minus baseline)
True
Predicted
daisy
dandelion
roses
sunflowers
tulips
-0.01 0.01 0.00 0.00 0.00
0.01 -0.01 0.00 0.00 -0.01
0.00 -0.01 -0.02 0.02 0.01
0.00 -0.01 -0.02 0.00 0.03
0.01 0.00 -0.01 0.00 0.00
0.02
0.01
0.00
−0.01
−0.02

26. Qualitative Analysis of Attention

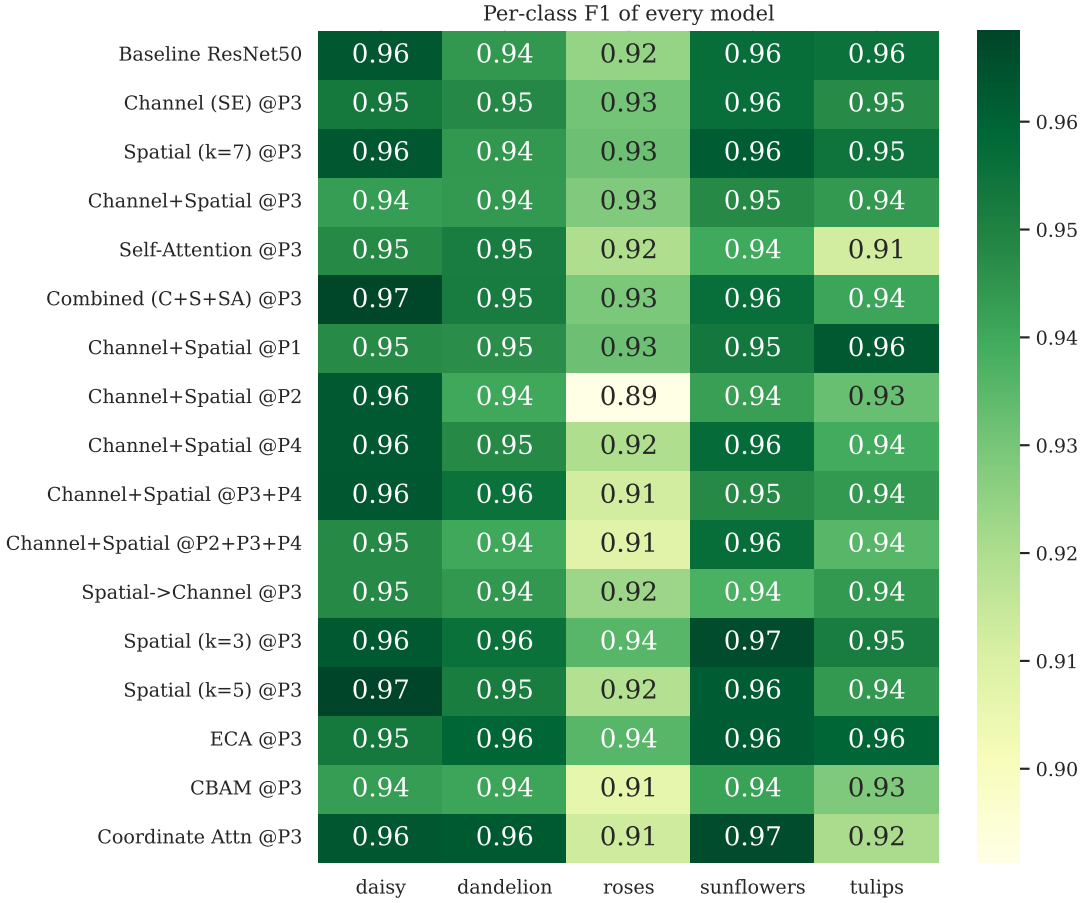
Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.



27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.
Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

Grad-CAM: Input | Baseline | Channel+Spatial attention

True: tulips (correct)



Baseline ResNet50
pred tulips (1.00)



Channel+Spatial
pred tulips (1.00)



True: daisy (correct)



Baseline ResNet50
pred daisy (1.00)



Channel+Spatial
pred daisy (1.00)



True: dandelion (correct)



Baseline ResNet50
pred dandelion (1.00)



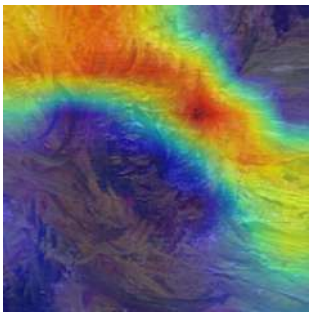
Channel+Spatial
pred dandelion (1.00)



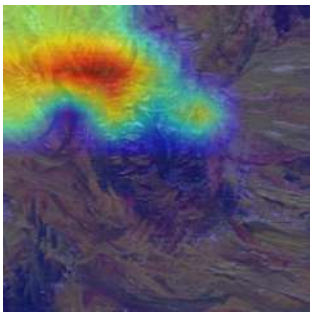
True: sunflowers (misclassified)



Baseline ResNet50
pred dandelion (0.57)



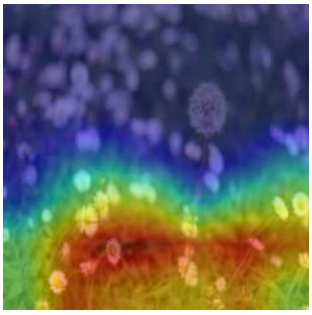
Channel+Spatial
pred dandelion (0.45)



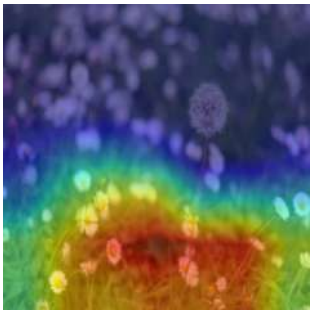
True: dandelion (misclassified)



Baseline ResNet50
pred daisy (0.99)



Channel+Spatial
pred daisy (1.00)



28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

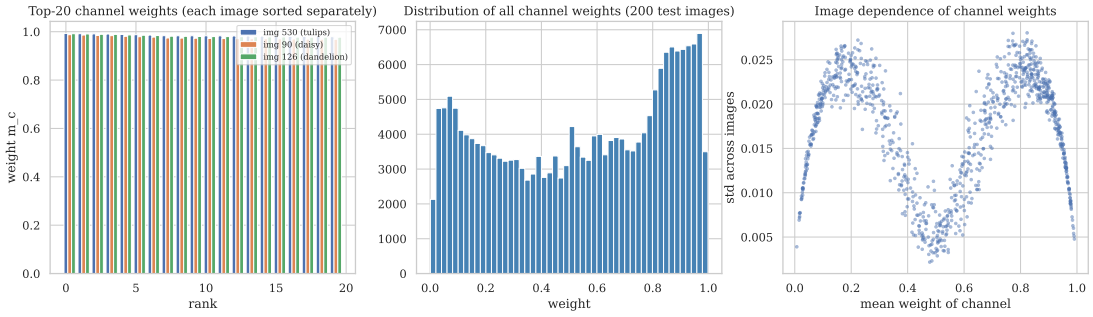
$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.



28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

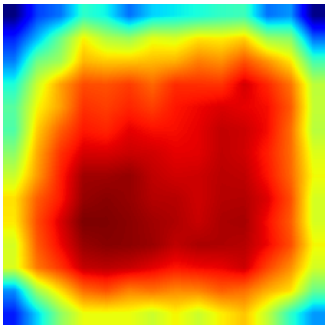
Students should identify the image regions receiving high attention values.

Spatial attention at P3 - Spatial (k=7) @P3

tulips (correct)



M_s range [0.71,0.98]



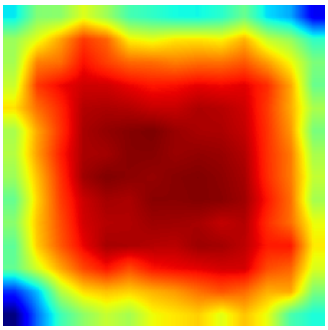
Overlay



daisy (correct)



M_s range [0.68,0.98]



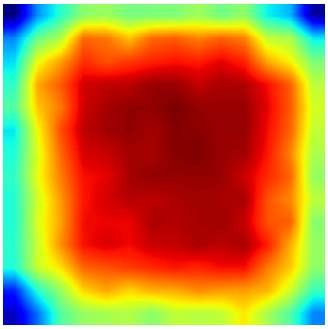
Overlay



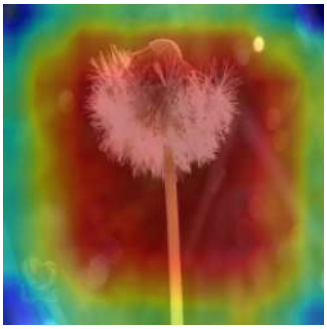
dandelion (correct)



M_s range [0.69,0.97]



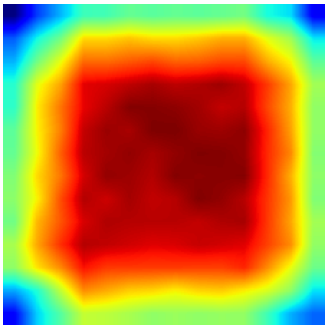
Overlay



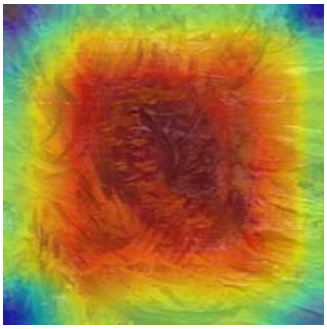
sunflowers (misclassified)



M_s range [0.65,0.93]



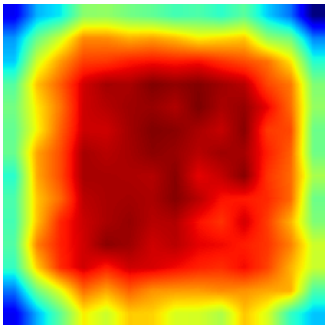
Overlay



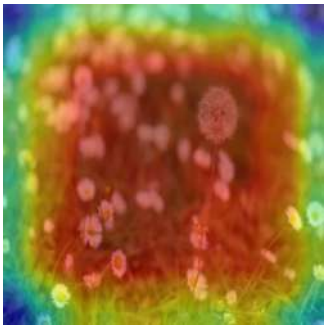
dandelion (misclassified)



M_s range [0.67,0.96]



Overlay

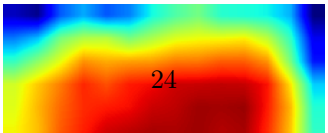


Spatial attention at P3 - Channel+Spatial @P3

tulips (correct)



M_s range [0.69,0.97]



Overlay

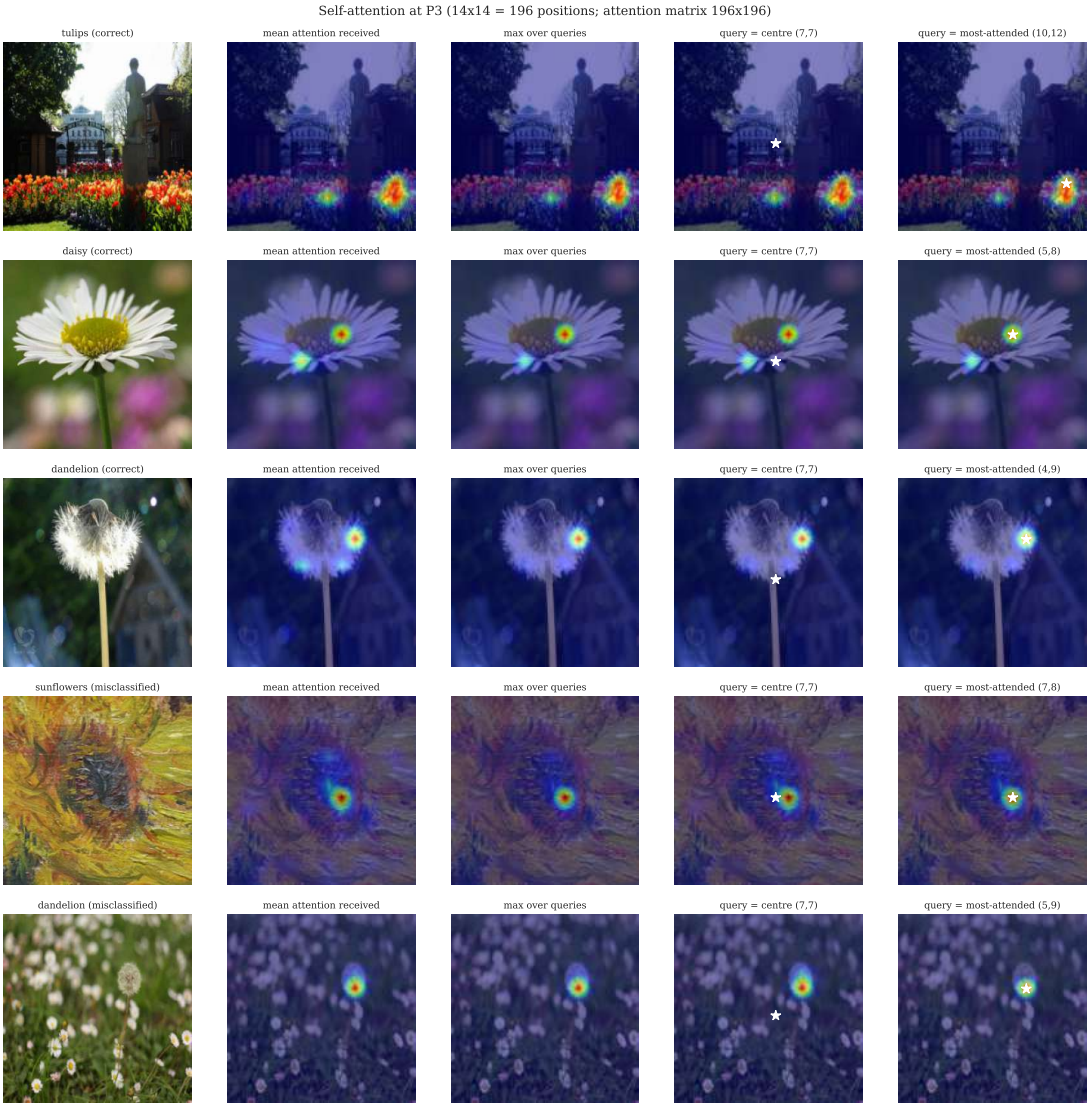


28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution. Students should analyze how distant regions interact within the feature map.



29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

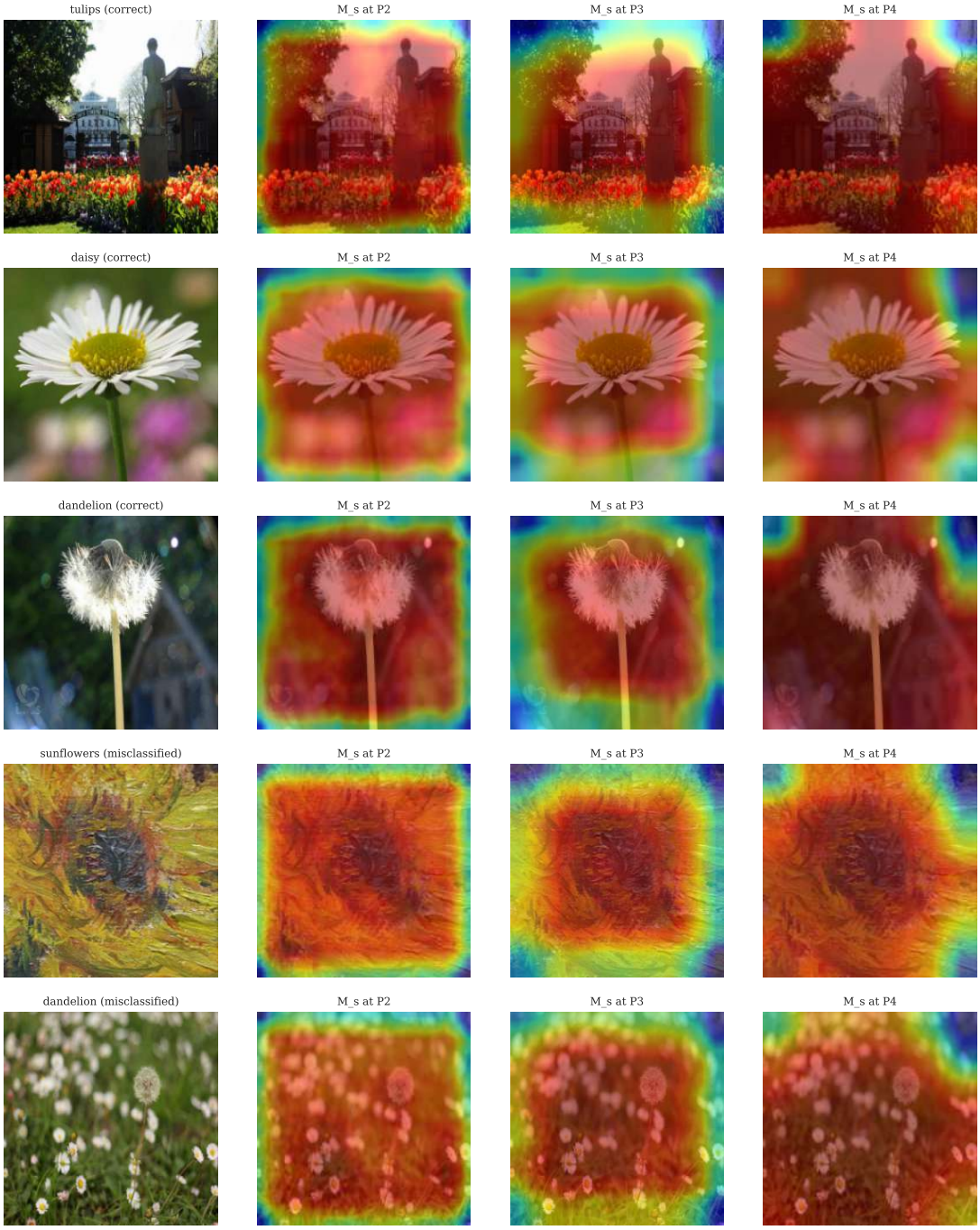
The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;

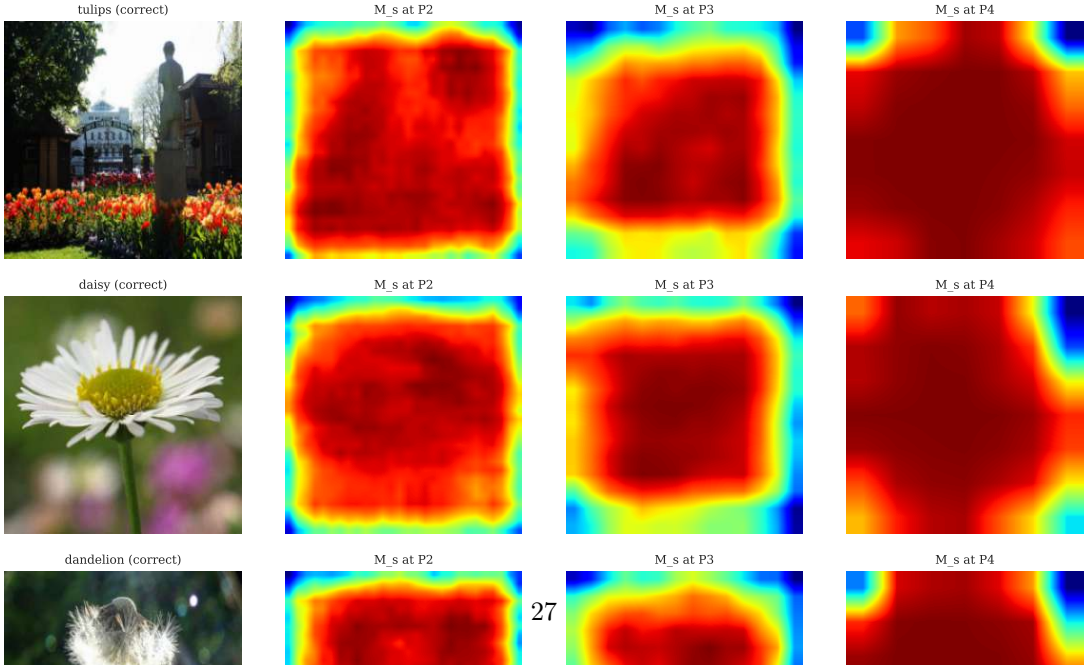
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

Spatial attention maps at different ResNet50 stages (overlay)



Spatial attention maps at different ResNet50 stages (heatmap)



30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

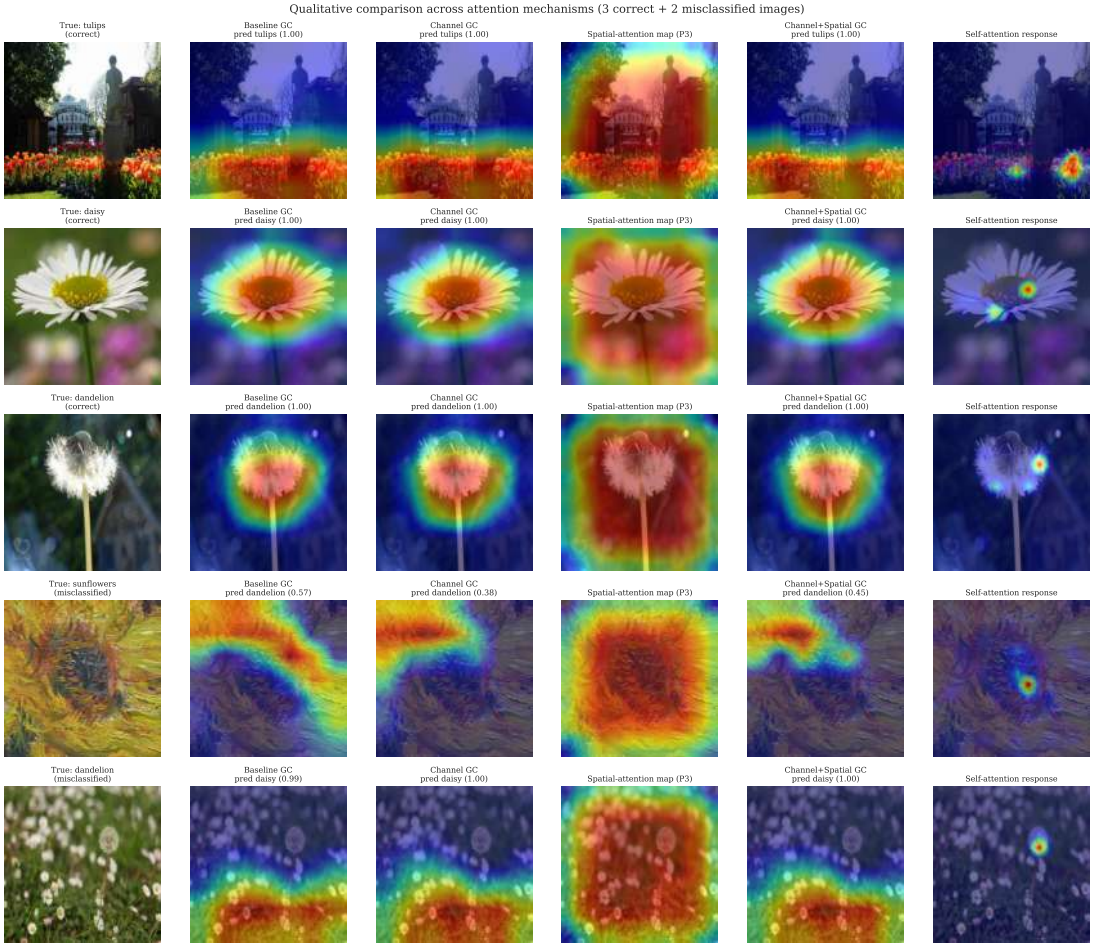
and

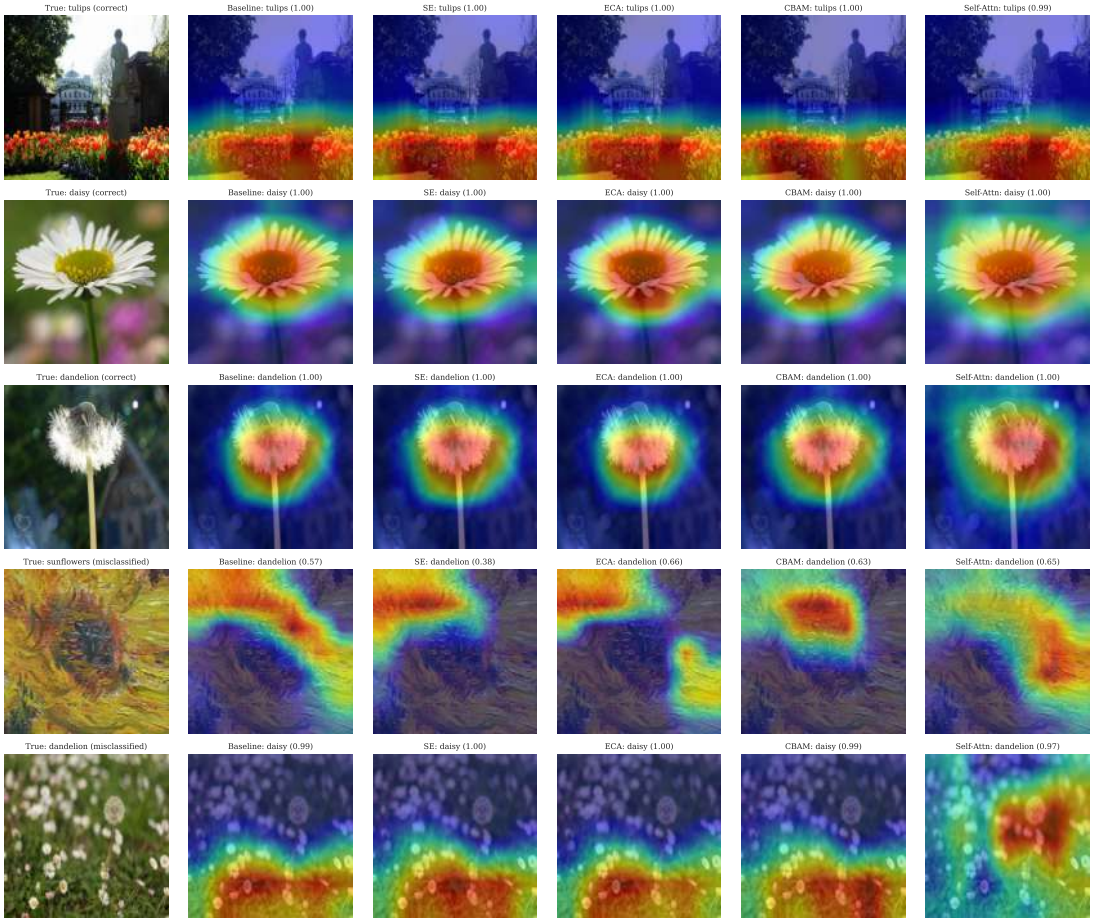
2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.


Qualitative comparison across attention mechanisms (3 correct + 2 misclassified images)
True: tulips (correct)
Baseline GC pred tulips (1.00)
Channel GC pred tulips (1.00)
Spatial-attention map (P3)
Channel+Spatial GC pred tulips (1.00)
Self-attention response
True: daisy (correct)
Baseline GC pred daisy (1.00)
Channel GC pred daisy (1.00)
Spatial-attention map (P3)
Channel+Spatial GC pred daisy (1.00)
Self-attention response
True: dandelion (correct)
Baseline GC pred dandelion (1.00)
Channel GC pred dandelion (1.00)
Spatial-attention map (P3)
Channel+Spatial GC pred dandelion (1.00)
Self-attention response
True: sunflowers (misclassified)
Baseline GC pred dandelion (0.57)
Channel GC pred dandelion (0.38)
Spatial-attention map (P3)
Channel+Spatial GC pred dandelion (0.45)
Self-attention response
True: dandelion (misclassified)
Baseline GC pred daisy (0.99)
Channel GC pred daisy (1.00)
Spatial-attention map (P3)
Channel+Spatial GC pred daisy (1.00)
Self-attention response


Grad-CAM: Input | Baseline | SE | ECA | CBAM | Self-Attention
True: tulips (correct)
Baseline: tulips (1.00)
SE: tulips (1.00)
ECA: tulips (1.00)
CBAM: tulips (1.00)
Self-Attn: tulips (0.99)
True: daisy (correct)
Baseline: daisy (1.00)
SE: daisy (1.00)
ECA: daisy (1.00)
CBAM: daisy (1.00)
Self-Attn: daisy (1.00)
True: dandelion (correct)
Baseline: dandelion (1.00)
SE: dandelion (1.00)
ECA: dandelion (1.00)
CBAM: dandelion (1.00)
Self-Attn: dandelion (1.00)
True: sunflowers (misclassified)
Baseline: dandelion (0.57)
SE: dandelion (0.38)
ECA: dandelion (0.66)
CBAM: dandelion (0.63)
Self-Attn: dandelion (0.65)
True: dandelion (misclassified)
Baseline: daisy (0.99)
SE: daisy (1.00)
ECA: daisy (1.00)
CBAM: daisy (0.99)
Self-Attn: dandelion (0.97)

30.1 Qualitative Criteria

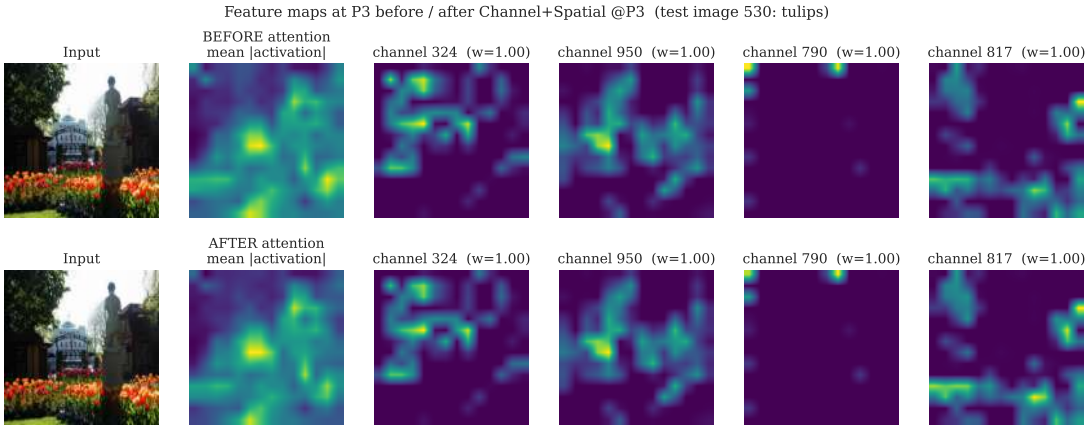
The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.



32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

$$\boxed{\text{Accuracy} + \text{Precision} + \text{Recall} + \text{Macro F1}}$$

33.2 Complexity Analysis

Report:

$$\boxed{\text{Parameters} + \text{Inference Time}}$$

33.3 Qualitative Analysis

Report:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.

5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

Attention Type

and

Attention Position.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. What is the significance of the residual connections in the ResNet50?
   Residual connections provide shortcuts which aid in gradient flow in the deeper networks, making optimization easy and solving the problem of degradation.
2. What is the difference between the conv2_x and conv5_x stages?
   conv2_x is an early stage having high spatial resolution and low-level features, whereas conv5_x is a deep stage having low spatial resolution and semantically rich information.
3. Why does the spatial resolution decrease with increasing network depth?
   Down-sampling saves computation and increases receptive field. Deeper layers make broader representations of the combined local information.
4. Why does the number of channels increase with network depth?
   By reducing the spatial dimensions, increasing number of channels enables the network to extract a large variety of patterns and semantically rich information.
5. What do you understand by low-level, intermediate and high-level features?
   Low-level features are about edges and color information, intermediate features are about texture and parts of objects, and high-level features are object-level patterns used in classification.
6. Purpose of attention mechanism
   Attention re-weights information in the features such that the network could focus on the informative ones and disregard the less important features.
7. What is the goal of channel attention?
   Channel attention seeks to learn the importance weight for each feature channel, thus emphasizing more informative channels with respect to the current input sample.

8. What is the goal of spatial attention?
   Spatial attention seeks to learn the importance weight for each spatial location, thereby emphasizing more informative parts of the image.
9. What is the difference between channel attention and spatial attention?
   Channel attention works across feature channels while spatial attention works across height and width dimensions; hence, they emphasize different properties of the feature map.
10. Why channel and spatial attention can be performed sequentially?
    First, channel attention could select informative features, followed by spatial attention highlighting the informative regions of the features. The two operations complement each other.
11. Why channel attention could perform differently for conv3_x and conv5_x?
    Earlier layers have rich textures and boundaries, but later layers have semantic features and fewer spatial locations. Channel weights thus highlight different aspects of data based on their depths.
12. Why is spatial attention helpful at an intermediate stage of ResNet?
    At the intermediate stage, there is still enough spatial information in addition to object components. Spatial attention highlights those components that will be downsampled away later.
13. Why is self-attention computationally costly if the feature map has a high spatial resolution?
    This approach considers spatial positions compared to each other. Therefore, its attention matrix is quadratic relative to the number of spatial positions.
14. What is the effect of spatial resolution growth on the attention matrix?
    Attention matrix has NxN elements for the feature map with N spatial positions. If each spatial dimension doubles, then the number of positions and matrix size quadruple and sixteenfold accordingly.
15. How does the effectiveness of attention depend on the insertion position?
    Channel-to-spatial attention in the position ablation achieved 94.92% at P1, 93.47% at P2, 94.19% at P3, and 94.56% at P4. The best among these tested positions was P1.
16. What is the difference between the attention in P_2 and the attention in P_4 regarding visualization?
    The P_2 feature map has a 56×56 resolution and is able to capture finer edges and texture, whereas the P_4 feature map has a 7×7 resolution, representing larger and more semantic features, and the visualizations present these different attention maps.
17. Does applying the attention mechanism in multiple layers always lead to better classification performance?
    Not necessarily. The model with only P_3 achieved an accuracy of 94.19%, whereas P3+P4 got an accuracy of 94.56%. Including P_2 further brought the accuracy down to 93.83%.
18. Does the behavior of the Channel→Spatial attention mechanism differ from that of the Spatial→Channel attention mechanism?
    Yes. The Channel→Spatial attention achieved an accuracy of 94.19%, and Spatial→Channel got 94.01%.
19. What effect does the kernel size of the spatial-attention operation have on the resulting attention map?
    The kernel size influences the spatial context that is used in attention calculation. In the presented results, for k=3 accuracy was 95.46%, k=5 was 94.92%, and k=7 was 95.10%.
20. What is the difference between SE and ECA attention?
    SE employs global channel descriptors followed by a small learnable gating mechanism. ECA makes use of local cross-channel interaction without reducing dimensionality and introduces few extra parameters.
21. How is CBAM attention different from SE and ECA attentions?
    CBAM combines channel attention with spatial attention. It uses the obtained channel and spatial attention weights sequentially to modify the feature tensor.
22. How is self-attention different from channel attention?
    Self-attention correlates spatial locations with each other to model dependencies between distant locations. In contrast, channel attention assigns weights to channels but does not create pairs of all possible spatial locations.
23. Does the attention with the highest number of parameters necessarily have the highest accuracy?
    Not necessarily. ECA yielded the highest accuracy (95.46%) with five additional attention parameters, while self-attention added 1,312,513 attention parameters and produced 93.47% accuracy.
24. What is the difference between internal attention map and Grad-CAM?
    An internal attention map is generated by the attention module directly in the forward pass. Grad-CAM relies on gradients of target class scores to identify important convolutional areas.
25. Why is Grad-CAM considered a post-hoc explanation method?
    Grad-CAM is a post-hoc method as it is calculated after training the model and generating its prediction; it explains this prediction without modifying the model in any way.
26. Is the attention model suppressing the background in irrelevant places?
    While the spatial maps and Grad-CAM overlays can help see what areas are activated by the models, it does not prove that all irrelevant background is indeed suppressed in all images.
27. Do the attention maps of correctly classified and incorrectly classified images look different?
    Yes, the visual comparison may vary between correct and incorrect predictions as the latter might demonstrate either diffuse activation or highlighting of less discriminative regions.
28. For which flower classes does the attention work better?

For class-wise comparison, roses gained small F1 improvement of about 0.003, whereas daisy had the biggest drop of around 0.020 for Channel+Spatial at P3 compared to the baseline.

29. What kind of information is provided by the confusion matrix that cannot be derived just from the overall accuracy?
    The confusion matrix indicates the classes that are confusing each other and whether errors happen in specific classes, information that a single number of overall accuracy cannot show.
30. Why do we need macro F1 besides the overall accuracy?
    Since the macro F1 score treats all classes equally by taking an average of class-wise F1 scores, it is important when performances vary between classes.
31. What does the term accuracy–complexity trade-off mean?
    The accuracy-complexity trade-off is the relationship between classification performance and parameters needed for computation. A small accuracy difference would not justify a more complex model.
32. What does the term F1–inference-time trade-off mean?
    The F1-inference-time trade-off is the comparison of balanced class performance against the time it takes to make a prediction. This is useful in finding out whether an improvement in the model's F1 performance is justified by the increased inference time.
33. Why does every ablation study need to have the same dataset split?
    It ensures that both models get evaluated using the same examples. Otherwise, any differences in scores will come from the different data used instead of the attention mechanism.
34. Why does the same training protocol have to be used for the comparison of different attention mechanisms?
    This ensures that any changes in the model's performance will result solely from the attention mechanism comparison.
35. What conclusions can be drawn from the Grad-CAM experiment?
    The comparison of the Grad-CAM experiments shows the activation distribution for the classes in question in regard to the input. These maps need to be analyzed in combination with the prediction and input.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**

10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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